

Class 08: Breast Cancer Mini Project

Barry (PID: 911)

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a biopsy data-set from fine needle aspiration (FNA) of breast mass.

Data Import

The data is made available as a CSV file for download. We can read this in using `read.csv()`:

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names = 1)
```

```
head( wisc.df, 3 )
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.8	1001
842517	M	20.57	17.77	132.9	1326
84300903	M	19.69	21.25	130.0	1203
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	

842302	0.11840	0.27760	0.3001	0.14710	
842517	0.08474	0.07864	0.0869	0.07017	
84300903	0.10960	0.15990	0.1974	0.12790	
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003	0.006193	25.38	17.33	
842517	0.01389	0.003532	24.99	23.41	
84300903	0.02250	0.004571	23.57	25.53	
	perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302	184.6	2019	0.1622	0.6656	
842517	158.8	1956	0.1238	0.1866	
84300903	152.5	1709	0.1444	0.4245	
	concavity_worst	concave.points_worst	symmetry_worst		
842302	0.7119	0.2654	0.4601		
842517	0.2416	0.1860	0.2750		
84300903	0.4504	0.2430	0.3613		
	fractal_dimension_worst				
842302	0.11890				
842517	0.08902				
84300903	0.08758				

Make sure we remove or exclude the `diagnosis` column from the data-set that we use for further analysis - this is the expert diagnosis as either M or B:

```
wisc.data <- wisc.df[,-1]
diagnosis <- as.factor( wisc.df$diagnosis )
```

Exploratory data analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
  B   M  
357 212
```

```
sum( wisc.df$diagnosis == "M" )
```

```
[1] 212
```

Q3. How many variables/features in the data are suffixed with `_mean`?

We can use the `grep()` function to help us here:

```
colnames(wisc.data)
```

```
[1] "radius_mean"           "texture_mean"  
[3] "perimeter_mean"       "area_mean"  
[5] "smoothness_mean"      "compactness_mean"  
[7] "concavity_mean"       "concave.points_mean"  
[9] "symmetry_mean"        "fractal_dimension_mean"  
[11] "radius_se"            "texture_se"  
[13] "perimeter_se"         "area_se"  
[15] "smoothness_se"        "compactness_se"  
[17] "concavity_se"         "concave.points_se"  
[19] "symmetry_se"          "fractal_dimension_se"  
[21] "radius_worst"         "texture_worst"  
[23] "perimeter_worst"      "area_worst"  
[25] "smoothness_worst"     "compactness_worst"  
[27] "concavity_worst"      "concave.points_worst"  
[29] "symmetry_worst"       "fractal_dimension_worst"
```

```
length( grep("_mean", colnames(wisc.data), value = T) )
```

```
[1] 10
```

Principal Component Analysis (PCA)

We need to scale our data before PCA with the `scale=TRUE` argument to `prcomp()`.

```
wisc.pr <- prcomp(wisc.data, scale=TRUE)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010

	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335

	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966

	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997

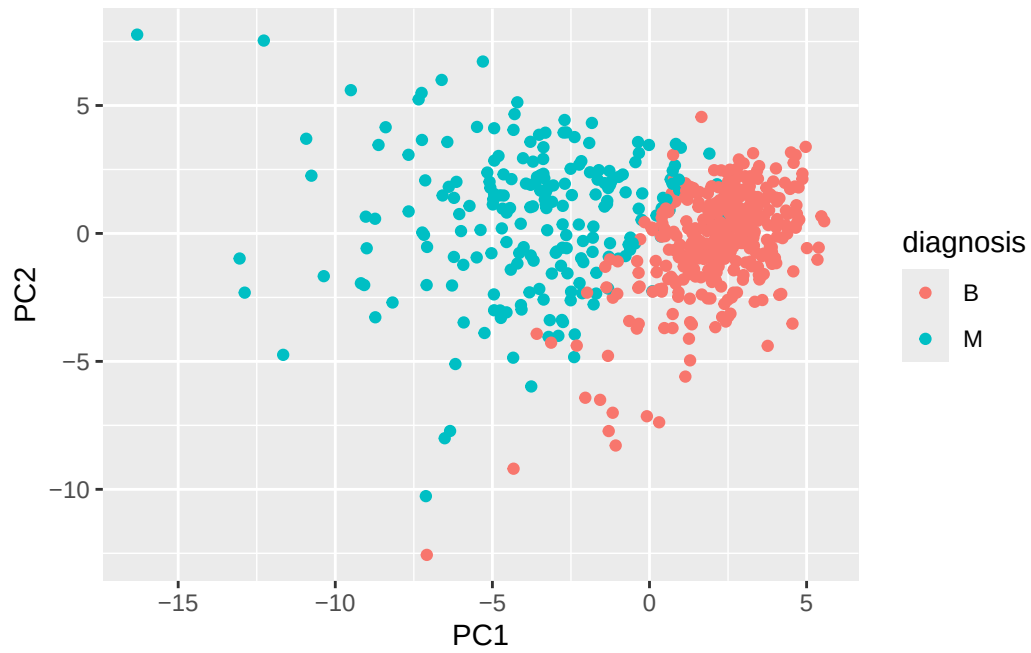
	PC29	PC30
Standard deviation	0.02736	0.01153
Proportion of Variance	0.00002	0.00000
Cumulative Proportion	1.00000	1.00000

Let's see the "PC score plot"

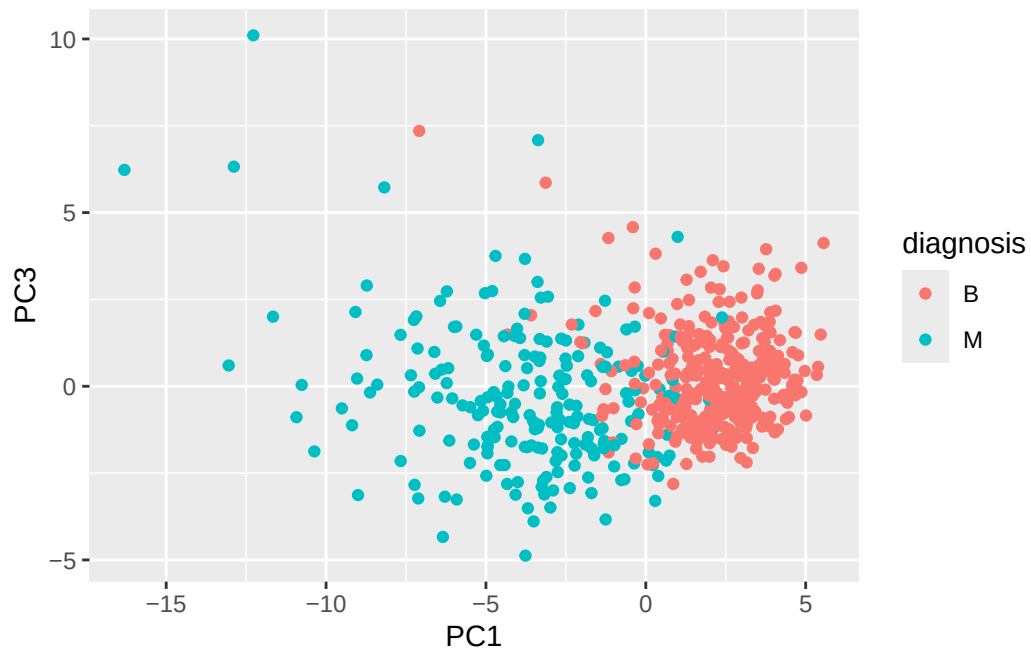
```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.4.3

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



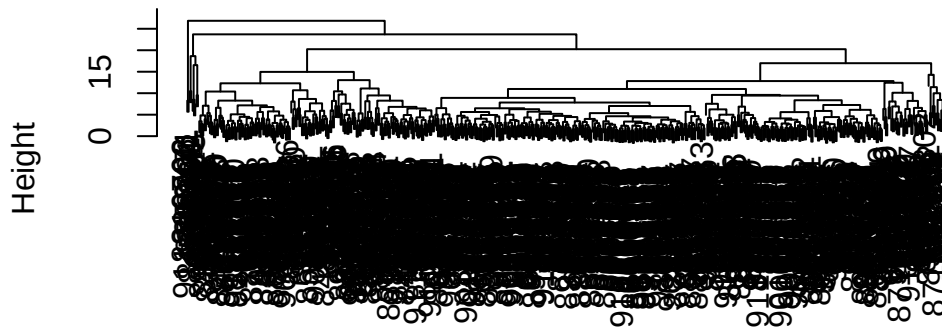
```
ggplot(wisc.pr$x) +  
  aes(PC1, PC3, col=diagnosis) +  
  geom_point()
```



Hierarchical clustering

```
wisc.hclust <- hclust( dist( scale(wisc.data) ) )  
plot(wisc.hclust)
```

Cluster Dendrogram

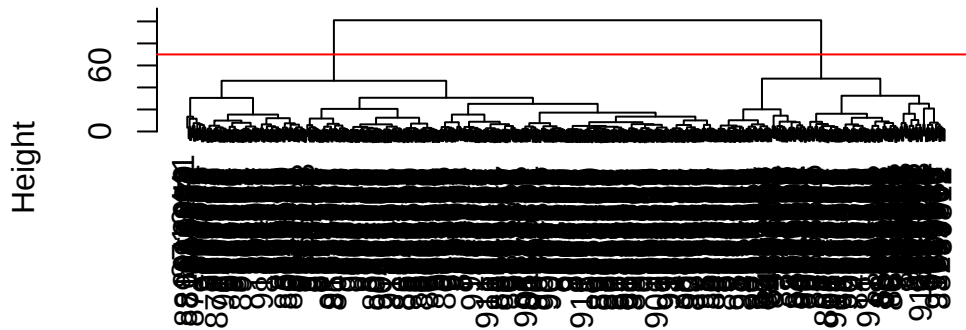


```
dist(scale(wisc.data))  
hclust (*, "complete")
```

Combining methods

```
d <- dist(wisc.pr$x[,1:4])  
wisc.pr.hclust <- hclust(d, method="ward.D2")  
plot(wisc.pr.hclust)  
abline(h=70, col="red")
```

Cluster Dendrogram



d
hclust (*, "ward.D2")

```
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1  2
171 398
```

How does this clustering `grps` correspond to the expert diagnosis?

```
table(diagnosis)
```

```
diagnosis
  B  M
357 212
```

```
table(diagnosis, grps)
```

```
      grps
diagnosis  1  2
  B      6 351
  M     165  47
```